PED chain: conflict with a crossing pedestrian

Derivation runs downward, from the collision (lagging SPI) back to the earliest observable signals

LAGGING SPI: PED collision

SPI: runs with a collision with a pedestrian per 10^5 manoeuvres < T_{PED}
Metric: contact between the ego vehicle and a pedestrian.

why

PROXIMATE CAUSE

The ego vehicle entered the crossing of the exit leg while a pedestrian was on it, and no stop was completed in time.

reaction tier (last resort, nearest the loss)

INTERMEDIATE LEADING SPI: AEB_PED

SPI: runs with an emergency braking event with a corridor pedestrian in conflict per 10^5 manoeuvres < T_{AEB_PED}
Metric: hard emergency brake while a pedestrian was near the closest point of approach.

decompose the proximate cause (kinematic precursor tier)

OR gate: either cause alone is sufficient

CAUSE A

The turn was committed without enough time to stop for a pedestrian whose path crosses the exit leg.
Planning subsystem.

CAUSE B

Occupancy of the crossing was not established before the manoeuvre was committed.
Perception subsystem.

SIGNAL A

Time to the closest point of approach between ego vehicle and pedestrian, gated on the miss distance.

SIGNAL B

Confidence of the crosswalk-occupancy estimate, sampled each frame.

observability and independence check

LEADING SPI: PED_CPA

SPI: runs with PED_CPA below θ per 10^5 manoeuvres < T_{PED_CPA}
Metric: minimum time to the closest point of approach.

NOT CARRIED FORWARD

Architectural tier: the signal is internal to the perception subsystem, which the simulation does not expose. Derived, but outside the empirical scope.

ATTACH TO THE SAFETY CASE

PED_CPA on G3.1 as leading evidence; AEB_PED as intermediate leading and the PED collision as lagging evidence on G3, the pedestrian-conflict goal. The perception branch corresponds to the undeveloped goal G3.2.

VALIDATE

Temporal precedence is checked against the PED collision only. Incidents are attributed by actor identity, so an oncoming or a rear impact is never counted here. The acceptance criterion for this chain is argued on minimum endogenous mortality, since a pedestrian neither selects the vehicle nor benefits from it.

AEB_PED attaches directly, independent of the kinematic tier